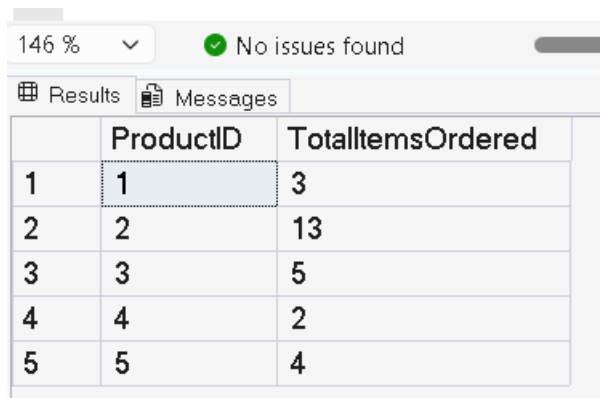


## GROUPING AND AGGREGATING DATA

- 1) Write a query that shows the total number of items ordered for each product. Use the Sales.SalesOrderDetail table.

```
SELECT  
ProductID,  
SUM(OrderQty) AS TotalItemsOrdered  
FROM Sales.SalesOrderDetail  
GROUP BY ProductID;
```

Output



The screenshot shows a SQL Server query results window. At the top, there is a zoom level of 146% and a status bar indicating 'No issues found'. Below the status bar, there are tabs for 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with two columns: 'ProductID' and 'TotalItemsOrdered'. The table contains five rows of data, numbered 1 through 5 in the first column.

|   | ProductID | TotalItemsOrdered |
|---|-----------|-------------------|
| 1 | 1         | 3                 |
| 2 | 2         | 13                |
| 3 | 3         | 5                 |
| 4 | 4         | 2                 |
| 5 | 5         | 4                 |

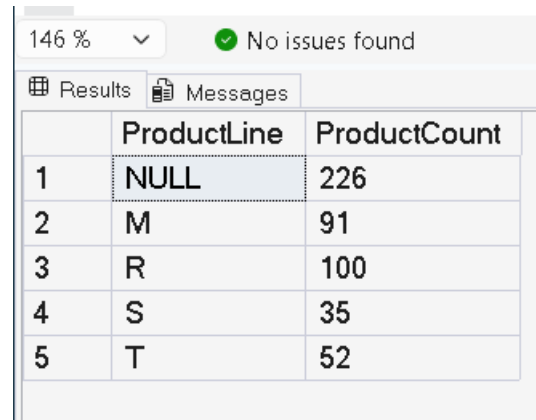
- 2) Write a query using the Production.Product table that lists the count of products in each ProductLine.

```
SELECT  
ProductLine,  
COUNT(*) AS ProductCount
```

FROM Production.Product

GROUP BY ProductLine;

## Output



|   | ProductLine | ProductCount |
|---|-------------|--------------|
| 1 | NULL        | 226          |
| 2 | M           | 91           |
| 3 | R           | 100          |
| 4 | S           | 35           |
| 5 | T           | 52           |

**3) Write a query that creates a sum of the LineTotal in the Sales.SalesOrderDetail table grouped by the SalesOrderID. Include only those rows where the sum exceeds 1000.**

SELECT

SalesOrderID,

SUM(LineTotal) AS TotalAmount

FROM Sales.SalesOrderDetail

GROUP BY SalesOrderID

HAVING SUM(LineTotal) > 1000;

## Output

| 146 %     | ▼            | ✔ No issues found |
|-----------|--------------|-------------------|
| 📊 Results | 📄 Messages   |                   |
|           | SalesOrderID | TotalAmount       |
| 1         | 1001         | 91500.00          |
| 2         | 1002         | 50000.00          |
| 3         | 1003         | 52000.00          |
| 4         | 1004         | 5000.00           |